

Question	Answer	Marks
2(a)(i)	mean/average square speed/velocity	B1
2(a)(ii)	$pV = NkT$ or $pV = nRT$	B1
	$\rho = Nm / V$ or $\rho = nN_A m / V$ and $k = nR / N$	B1
	$E_K = \frac{1}{2} m \langle c^2 \rangle$ with algebra to $(3/2)kT$	B1
2(b)(i)	no (external) work done or $\Delta U = q$ or $w = 0$	B1
	$q = N_A \times (3/2)k \times 1.0$	M1
	$N_A k = R$ so $q = (3/2)R$	A1
2(b)(ii)	specific heat capacity = $\{(3/2) \times R\} / 0.028$	C1
	= $450 \text{ J kg}^{-1} \text{ K}^{-1}$	A1

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Question	Answer	Marks
2(a)(i)	period = 6/2.5	01